XINGQIANG CHEN

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OBJECTIVE

Looking for **FULL TIME** roles in **AI Accelerator Compiler / System Software** · **LLM Inference Systems** · **HW/SW Co-design**.

INTERDISCIPLINARY BACKGROUND

Physics and biophysics training across computational chemistry, molecular dynamics, and computer science. Contests: **CUMCM National First Prize** (2012); **MCM Meritorious Winner** (2013); **Second Prize**, CCF Open Source GPU Innovation Ecosystem Regular Season S1, GPU Operator Optimization Challenge (2025).

EDUCATION

Doctoral Studies in Biophysics and Condensed Matter Sep. 2014 - Jul. 2018
Xiamen University, Department of Physics and Engineering; protein dynamics and QM/MM simulation
Certificate of completion; degree defense in progress

B.S. in Theoretical Physics Sep. 2010 - Jul. 2014
Xiamen University, Department of Physics and Engineering

SKILLS AND PROJECTS

Languages: C++, Python (primary); CUDA, Java, Scala, Go.

Compiler / Toolchain: MLIR, LLVM backends and ISA bring-up, TVM, graph partitioning, operator fusion, cost modeling, codegen, auto-tuning.

Chip Software / Verification: Palladium emulation, GDB co-debug against simulation models, firmware and weight-map generation, operator-level metrics (Cycles/MACs).

Inference Systems: Transformers, KV-Cache, quantization, batching and scheduling, Triton kernels, FlashAttention, CUPTI/Perfetto profiling.

Platforms: PyTorch, TensorFlow, ONNX, Kubernetes; Spark, Flink, Hadoop.

Open Source & Releases (`github.com/chenxingqiang` unless noted):

- **TraceSmith** — GPU profiling & replay across CUDA/ROCm/Metal/MACA/Ascend; C++ core, Python bindings, on PyPI. `TraceSmith`
- **YiRage Persistent Kernel** — LLM megakernel compiler and runtime, multi-backend. `YiRage`
- **JAX-sklearn** — sklearn-compatible API on JAX; 13,058 tests passing; on PyPI. `jax-sklearn`
- **Secret-Learn** — Privacy-preserving ML: federated learning, secret sharing, split learning. `Secret-Learn`
- **YouTube-DNN-RecSys** — TensorFlow + Spark retrieval and ranking (60 stars). `YouTube-DNN-RecSys`
- **Research CLI** — Command-line AI tooling for research workflows. `iechor-research/research-cli`
- **Questions-Gen** — Math-problem generation (Qwen3-14B, GRPO, distillation); CLI and HF weights. `GenQuestion`

EXPERIENCE

Compiler & Toolchain Expert, Witmem Technology
Hangzhou, China
Compiler / Palladium Verification *Dec. 10, 2025 - Present*

- Owned the product-line compiler and Palladium emulation for a CIM chip: model compile, codegen, on-chip execution, and defect isolation on the emulator.
- Brought up software on Palladium: ran compiler output on the emulator and jointly debugged with GDB across the simulation model, silicon, and system software.

- Implemented **CIM-MLIR**: PyTorch/TensorFlow/ONNX lowered via TOSA/StableHLO to CPU, GPU, and CIM, covering frontend, graph partition, hardware backend, and runtime.
- Implemented cost-driven graph partition and CIMHigh/Mid/Low IR; used OR-Tools for Die/L2/L0 allocation; emitted burnable **config.bin**, weight **map.bin**, GPU PTX, and LLVM host code.
- Implemented a cross-device runtime: per-device executors, unified memory, compute/transfer overlap and triple buffering; C++/Python APIs with latency, utilization, and transfer-cost counters.
- Implemented **GRC-PIM**: drift-gated, GEMV-reusing residual calibration and soft-decision error correction for PIM; submitted to *npj Unconventional Computing* (see Publications).

Senior AI Compiler Engineer, Yizhu Tech Ltd.
Hangzhou, China
R&D Department *Sep. 2022 - Dec. 2025*

- Led **YiRage Persistent Kernel (YPK)**, compiling end-to-end LLM decode into a single megakernel; **1.2×–6.7×** end-to-end latency gain over a PyTorch eager baseline on Llama2-7B, single A100 (varies with batch and sequence length).
- Implemented fused operators (RMSNorm+Linear, GQA) and a persistent execution model, cutting kernel-launch and intermediate-tensor round trips; largest gains on the batch=1 decode path.
- Built the scheduling layer and integrated Triton and FlashInfer kernels; multi-GPU scaling validated up to **8** devices with NVSHMEM-backed tensors.
- Developed graph compilation and auto-tuning: kernel parameter search and KV-cache-friendly memory layout to reduce memory traffic.
- Filed **2** invention patents; delivered the stack into **2** datacenter customer PoCs.
- Owned the LLVM backend Target and complex-ISA support for the in-house SoC, driving backend standardization and the software ecosystem.
- Built model-conversion and operator-level evaluation: integrated TVM, ONNX, MLCommons, TensorFlow/PyTorch into **YZlite**, reporting Cycles/MACs and multi-chip benchmarks. `ReRAM-Labs/yzlite`

Privacy-preserving Algorithms Engineer, Trustbe
Hangzhou, China
IT Group *Jan. 2022 - Sep. 2022*

- Built a homomorphic-encryption federated learning platform for cross-institution modeling in finance, improving model quality over local-only baselines while keeping privacy constraints.
- Designed MPC protocols for joint training and inference of bank risk models, balancing protocol security against communication cost.
- Open-sourced **Secret-Learn**: sklearn-compatible API across federated learning, secret sharing, and split learning, JAX-accelerated and integrated with SecretFlow; on PyPI. `Secret-Learn`
- Filed **3** invention patents, all published: `CN114925786A`, `CN114997345A`, `CN115130681A`.

Staff Algorithms Engineer, BioMap
Beijing, China
Algorithm Engineering Group *Sep. 2021 - Dec. 2021*

- Built bioinformatics tooling and a unified data platform for large-scale sequence analysis, structure prediction, and functional annotation.
- Set up evaluation pipelines for protein structure prediction models (AlphaFold family) feeding downstream drug design.
- Two registered software copyrights (2021SR2004464, 2021SR1862397); open-sourced AlphaFold2 code walkthroughs (26 stars). `alphafold2-codecs`

Senior Algorithms Engineer, Yalla Group (NYSE: YALA)
Hangzhou, China
Data Center Engineering Group *Sep. 2020 - Sep. 2021*

- Rebuilt the ChatRooms recommender with multi-objective ranking; A/B tests showed significant gains in daily active time and 7-day retention (**42% → 61%**).
- Built Ludo risk control from real-time behavior features and ML models, materially reducing fraudulent refunds.
- Developed a card-game AI with deep RL for human-vs-bot play and matchmaking fill, improving off-peak match quality and retention.

Senior Algorithms Engineer, Gejia Network Tech — Hangzhou, China
Search & Recommendation Engineering Group — *Mar. 2019 - Sep. 2020*

- Designed multi-scenario ranking with collaborative filtering, deep ranking models, and a product knowledge graph; online experiments delivered double-digit relative CTR and CVR gains.
- Optimized the serving path with feature caching and candidate pruning, holding P99 latency under **50ms** through peak promotional traffic.
- Open-sourced YouTube-DNN-RecSys, a TensorFlow + Spark retrieval and ranking reference (60 stars). `chenxingqiang/YouTube-DNN-RecSys`

Machine Learning Engineer, Bianlifeng Tech — Beijing, China
Data Engineering Group — *Mar. 2017 - Mar. 2019*

- Built store-level sales forecasting across **50+** stores using gridded weather and spatiotemporal features, reducing stock-outs and dead stock while improving turnover.
- Linked customer identities to purchase history over tens of millions of events to power personalized recommendations.
- Shipped a product recognition model for **500+** SKUs used in fast checkout, cutting checkout and queue time.

RESEARCH & INTERNSHIPS

Molecular Dynamics Simulation of Proteins with QM/MM model — Xiamen university, China
(China National Funds) — *Sep. 2014 - Mar. 2017*

- Ran QM/MM protein dynamics simulations and conformational analysis; results published in Acta Cryst. D, PLoS ONE, and Phys. Rev. E (see Publications).

Liquid Biopsy ct-DNA Kit and DNA Methylation Analysis — Lepu Medical, China
(R&D Engineer Intern) — *Mar. 2016 - Mar. 2017*

- Analyzed roughly 10,000 public gene-chip samples (TCGA/GEO) to localize Septin9 methylation sites, an early colon-cancer marker, informing biomarker selection for a clinical diagnostic kit.

AWARDS & SERVICE

- **National First Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM), 2012.
- **Meritorious Winner**, Mathematical Contest in Modeling (MCM/COMAP), 2013.
- **6th** worldwide, International Solar Decathlon Competition, 2013.
- Ranks **57** and **32** of 1739 teams, IEEE ICDM Global Meteorological AI Challenge, 2018.
- **Second Prize**, 8th CCF Open Source Innovation Competition — Open Source GPU Innovation Ecosystem Regular Season S1, GPU Operator Optimization Challenge (CCF/ODTC), 2025. CCF announcement
- Judge, first screening, Best Practices of Applied Algorithms, 2021.

PUBLICATIONS

- Chen X. GRC-PIM: A Drift-Gated, GEMV-Reusing Residual Calibrator for Soft-Decision Error Correction in Processing-in-Memory. *npj Unconventional Computing* (submitted).
- Cai Y, Su M, Ahmad A, Hu X, Sang J, Kong L, Chen X, Wang C, Shuai J, Han A. Conformational dynamics of the essential sensor histidine kinase WalK. *Acta Crystallographica Section D: Structural Biology.* 2017 Oct 1;73(10):793-803.
- Xia H, Chen X, Chen X, Cheng J, Liu Y, Chen X, Zhang Q, Zou G. Chiral discrimination of amino acid enantiomers based on different interactions with Cu2+. *Sensors and Actuators B: Chemical.* 2018 Jan 1;254:44-51.
- Ahmad A, Cai Y, Chen X, Shuai J, Han A. Conformational dynamics of response regulator RegX3 from Mycobacterium tuberculosis. *PloS one.* 2015 Jul 22;10(7):e0133389.
- Chen Y, Qi H, Li X, Cai M, Chen X, Liu W, Shuai J. Suppressing effect of C a 2+ blips on puff amplitudes by inhibiting channels to prevent recovery. *Physical Review E.* 2016 Aug 15;94(2):022411.

Open Source Architecture Appendix

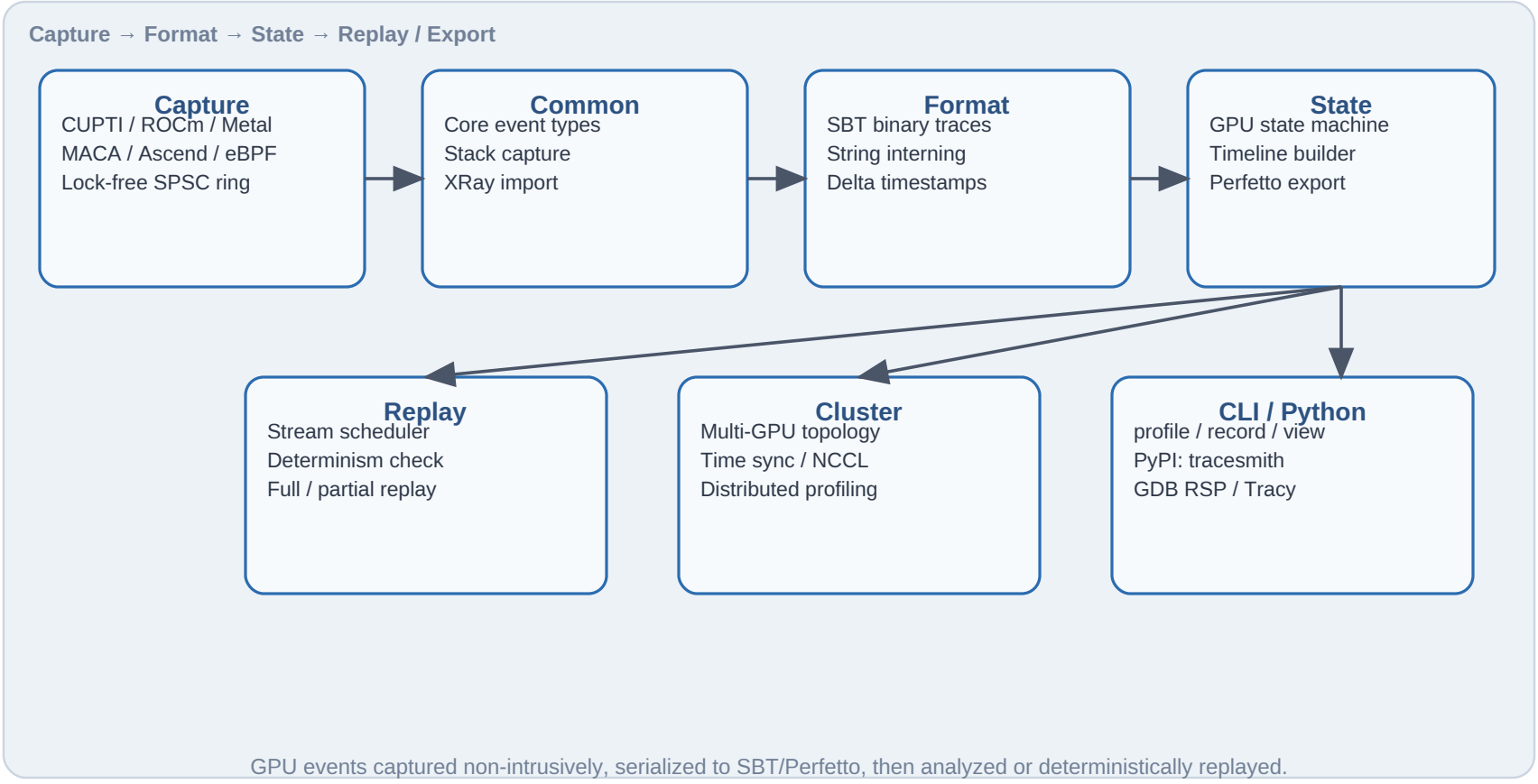
Chen Xingqiang · github.com/chenxingqiang

TraceSmith · YiRage · JAX-sklearn · Secret-Learn

YouTube-DNN-RecSys · Research CLI · Questions-Gen

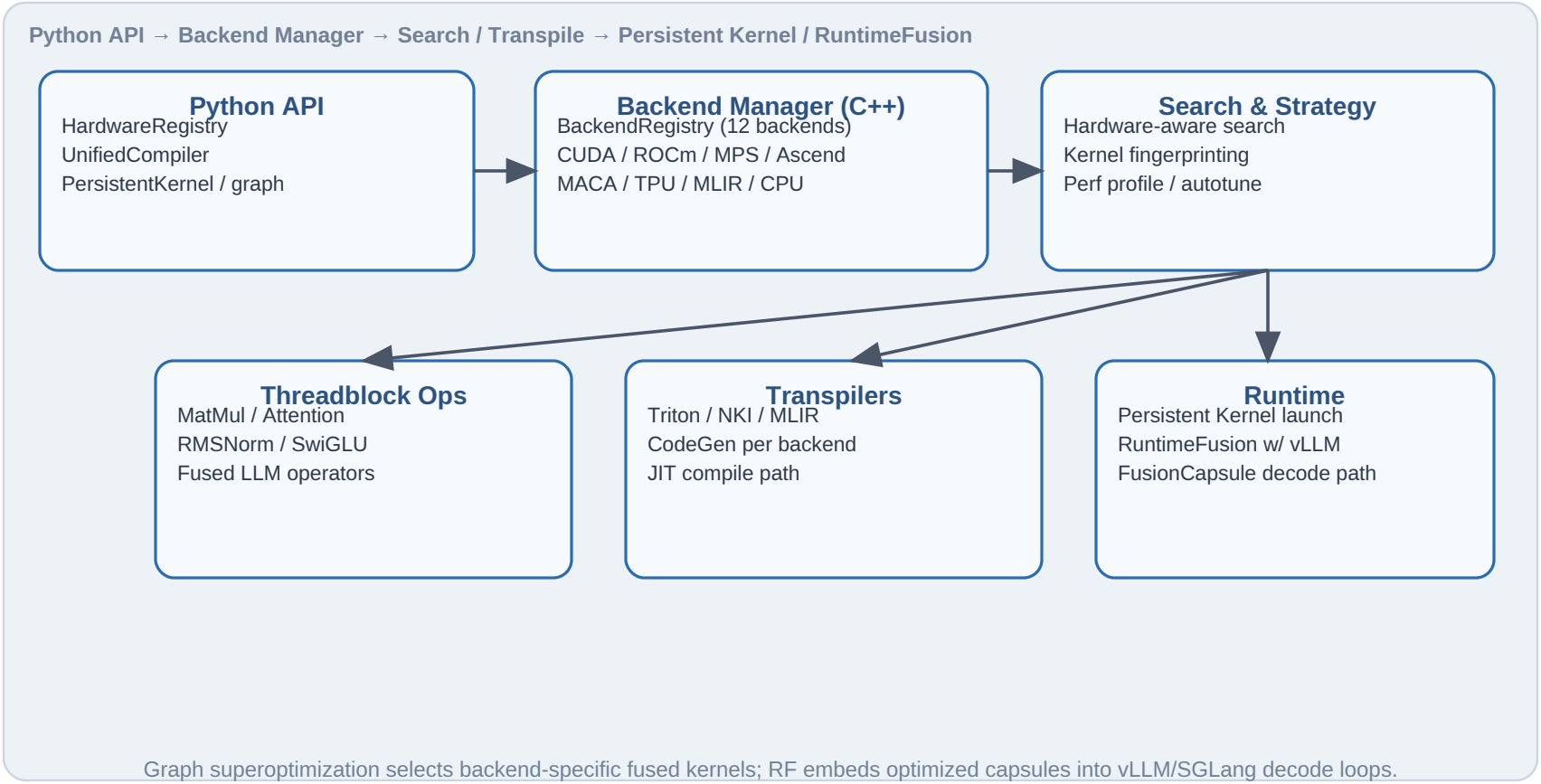
TraceSmith

Cross-platform GPU Profiling & Replay · github.com/chenxingqiang/TraceSmith



YiRage (YPK)

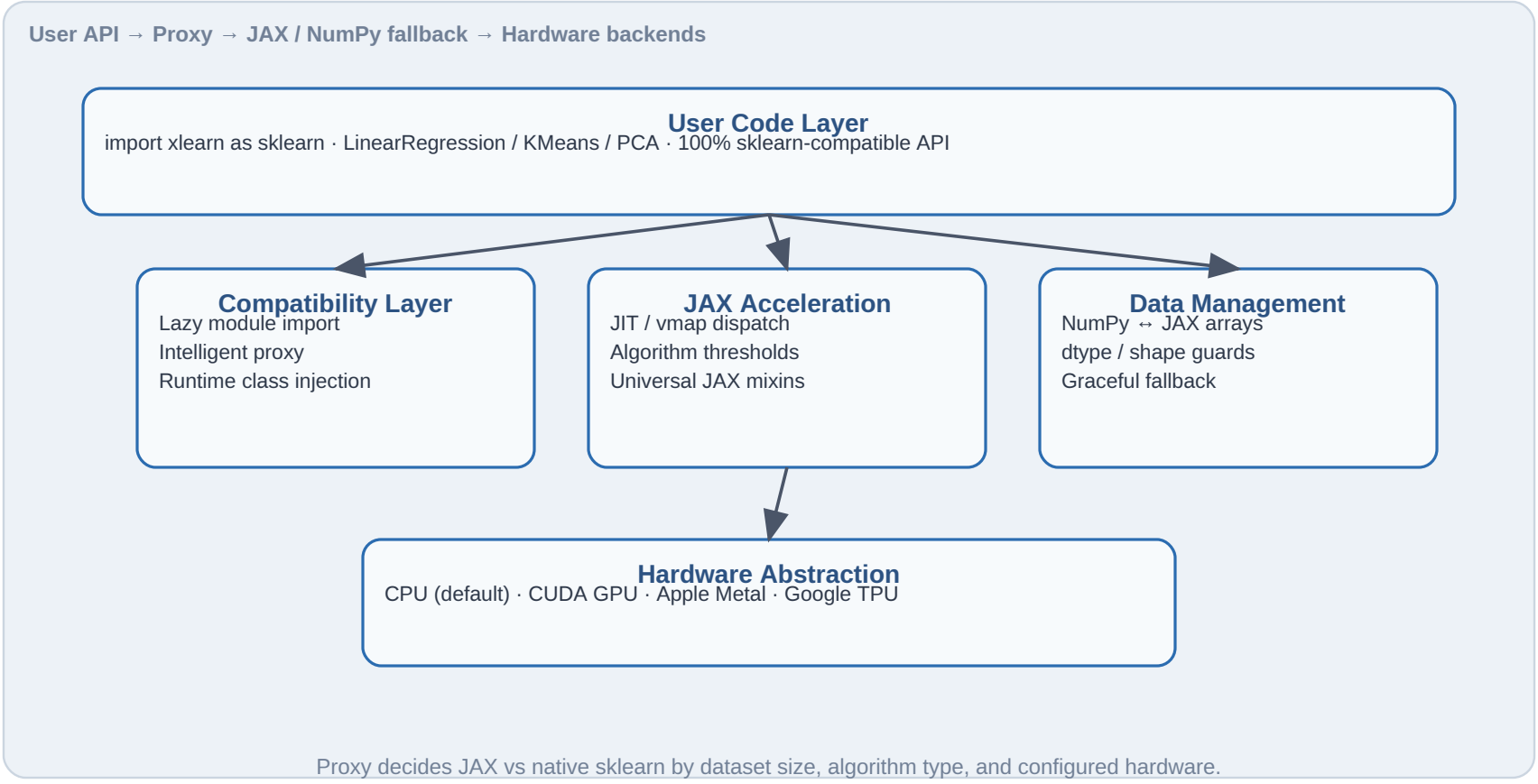
Multi-backend LLM megakernel compiler & runtime · github.com/chenxingqiang/YiRage



Graph superoptimization selects backend-specific fused kernels; RF embeds optimized capsules into vLLM/SGLang decode loops.

JAX-sklearn (xlearn)

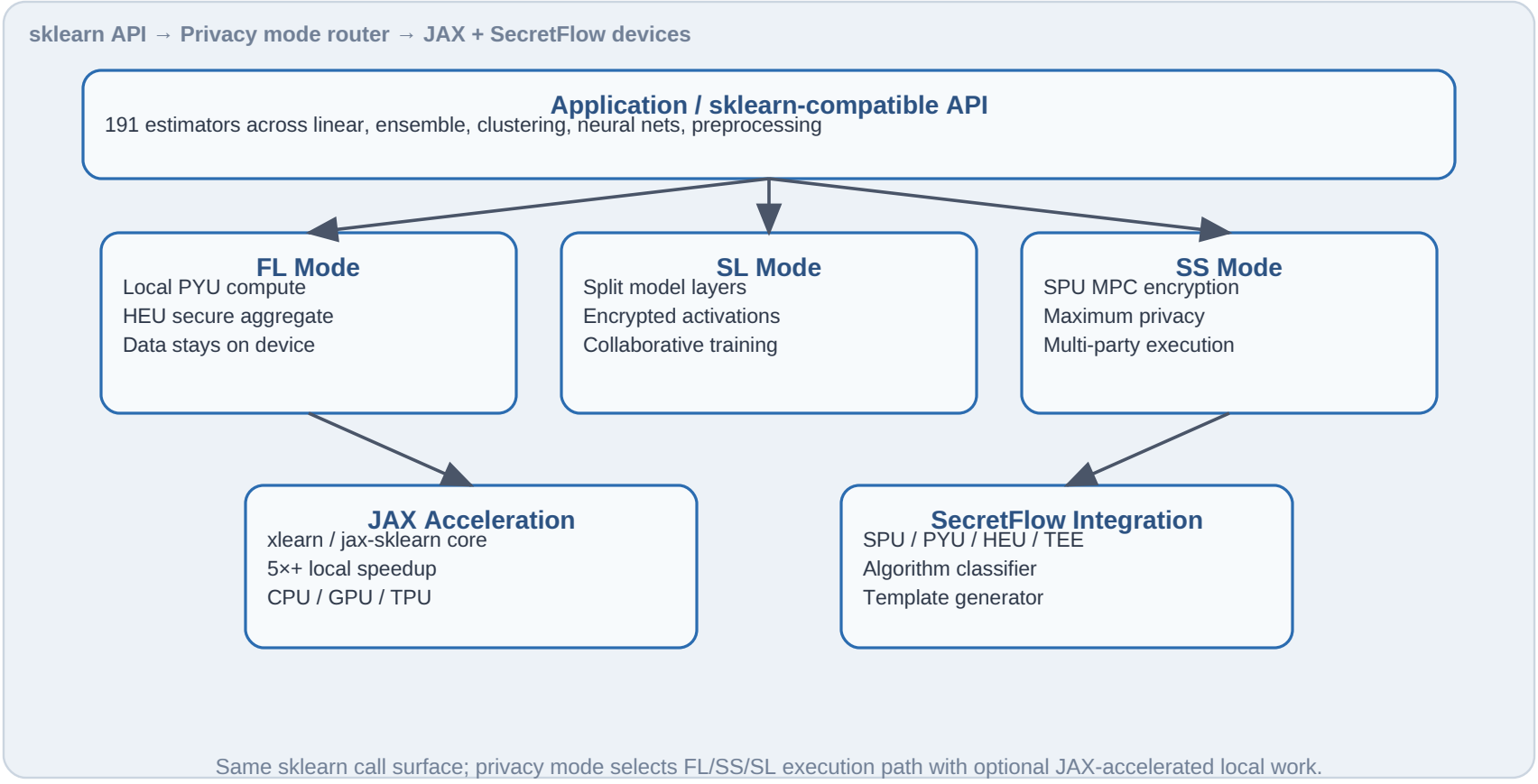
Drop-in sklearn API with automatic JAX acceleration · github.com/chenxingqiang/jax-sklearn



Proxy decides JAX vs native sklearn by dataset size, algorithm type, and configured hardware.

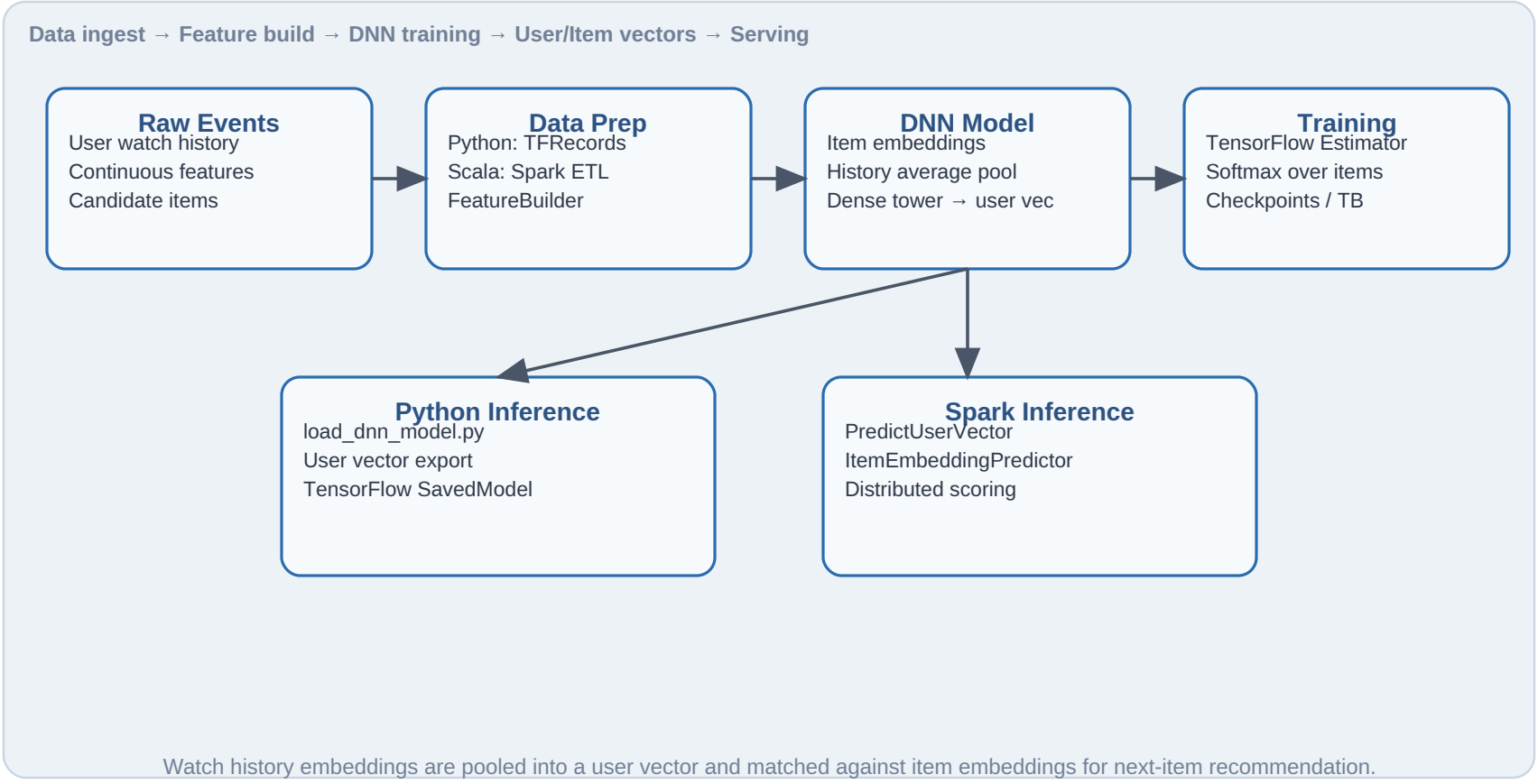
Secret-Learn

Privacy-preserving ML · 191 algorithms × FL/SS/SL · github.com/chenxingqiang/Secret-Learn



YouTube-DNN-RecSys

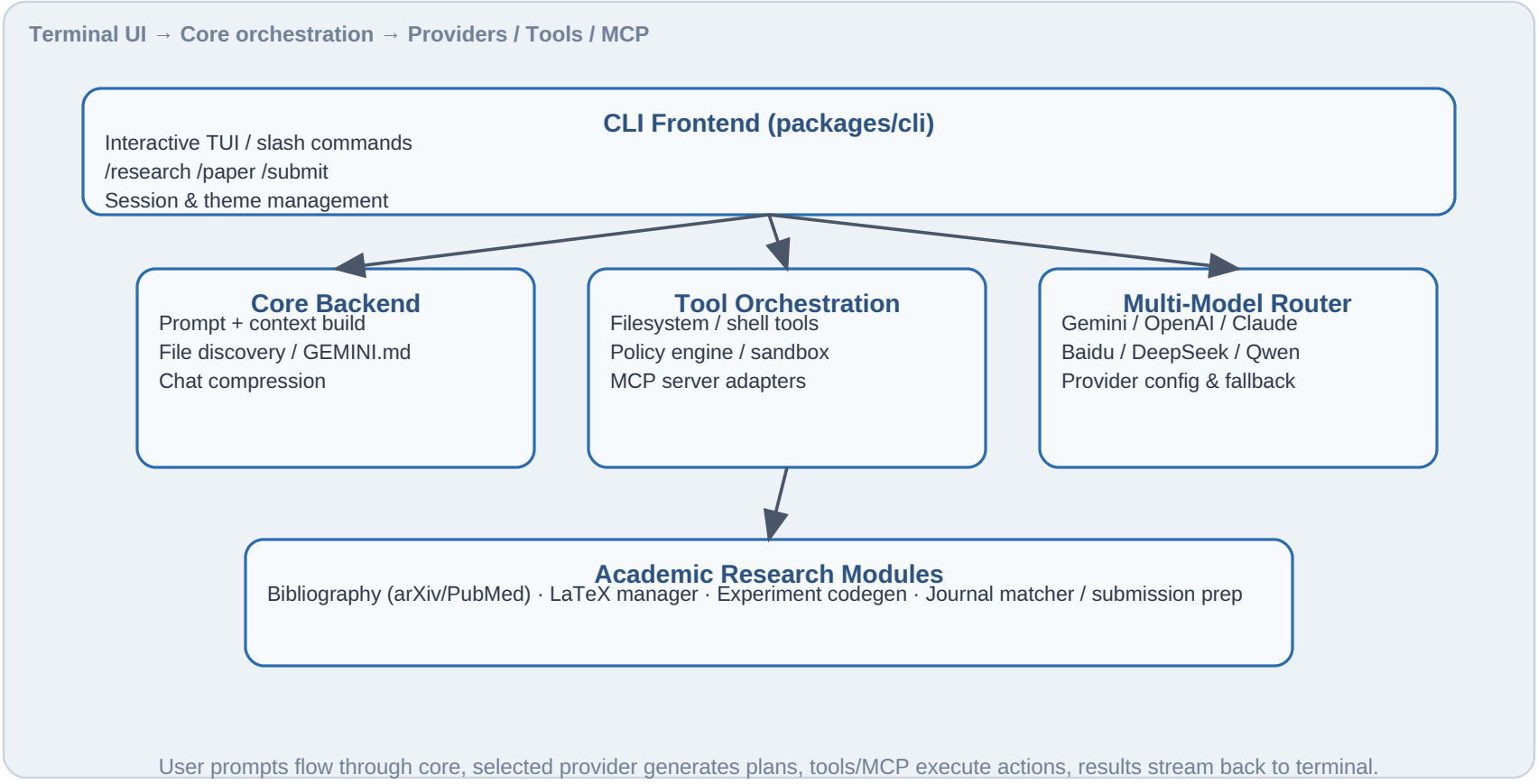
Two-tower recall pipeline · TensorFlow + Spark · github.com/chenxingqiang/YouTube-DNN-RecSys



Watch history embeddings are pooled into a user vector and matched against item embeddings for next-item recommendation.

Research CLI

CLI AI workflow + academic research tools · github.com/iechor-research/research-cli



Questions-Gen

Math competition problem generator · Qwen3-14B · github.com/chenxingqiang/GenQuestion

